

BMU–GW BMS gateway

Rev B0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Isolated CAN FD + RS–485 between CTRL–C2000 and the battery BMS

01 Isolated CAN + RS–485

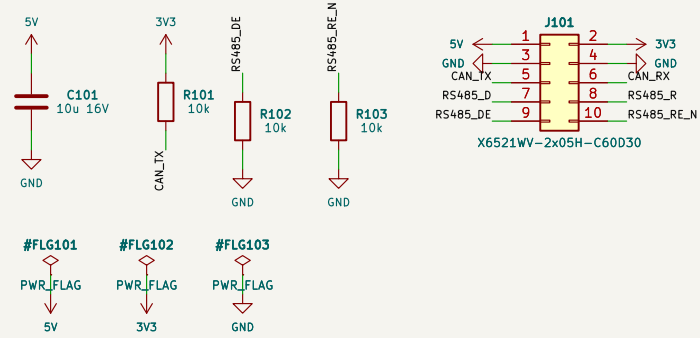
Host header, push–pull isolated 5 V, ISO1042 CAN FD, ISO1410 RS–485, TVS, terminations, field connector

File: sheets/01_gateway.kicad_sch

The field side (GND_BMS) floats with the BMS communication ground; only U–isolators and the bias transformer cross the barrier (checked on every build).
The isolated 5 V is unregulated (5 V x 1.1 minus one Schottky drop): 4.8–5.5 V over a 5 V +/-5 % input, inside the 4.5–5.5 V VCC2 window of ISO1042. R preload keeps it bounded at no load.

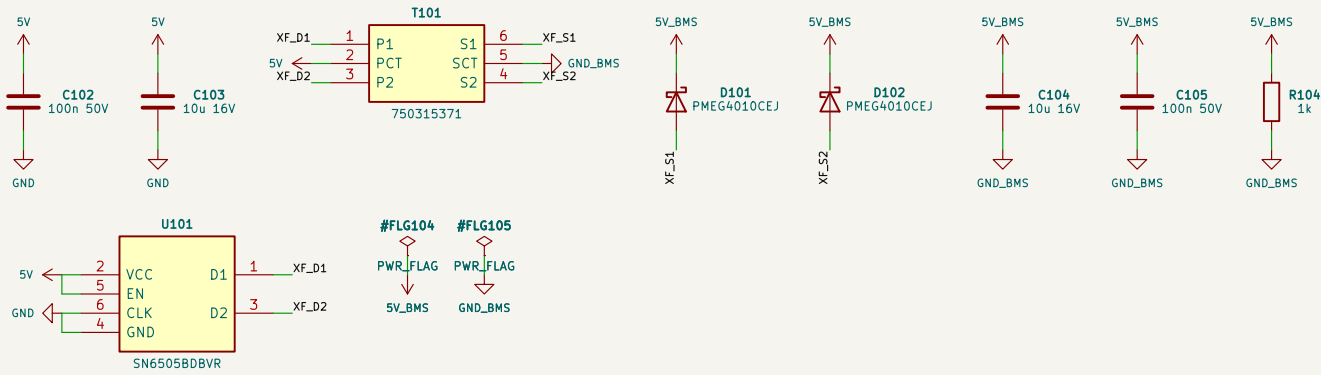
Not bench–validated. PELV = controller side, BMS = field side. Sheet 1 Schematics authored by Chinmoy Bhuyan Barrier: 5 kVrms reinforced isolators; bias transformer 2500 VAC test (functional/basic)		
Sheet: / File: BMU–GW.kicad_sch		
Title: BMU–GW BMS gateway – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: B0
KiCad E.D.A. 10.0.6	Id: 1/2	

HOST HEADER (PELV)



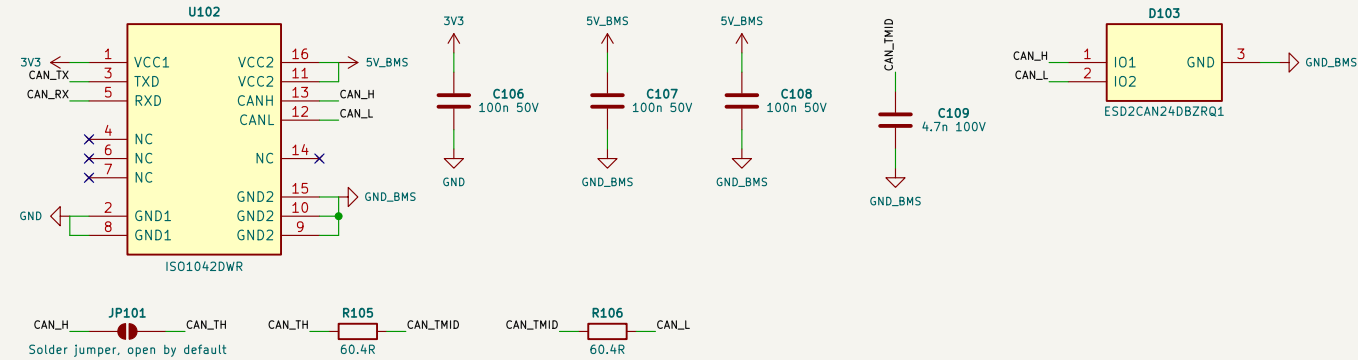
From CTRL-C2000 / SYS-IO-AUX.
DE and -RE pulled low: driver off, receiver on at reset.

ISOLATED 5 V BIAS



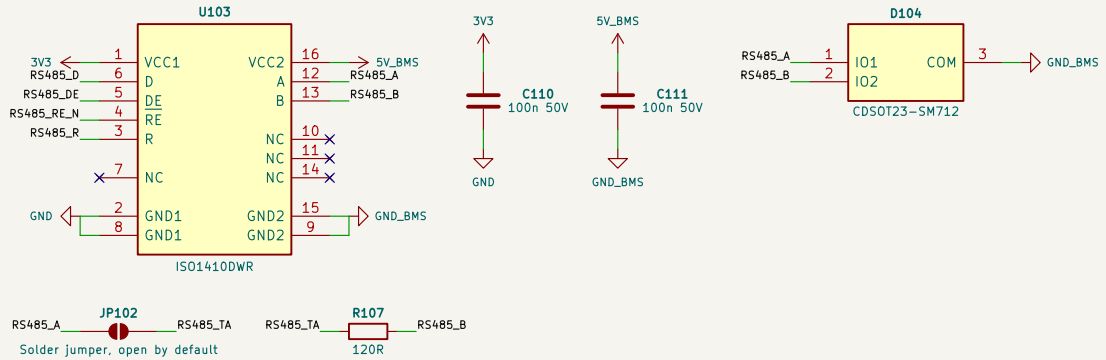
SN6505B push-pull, 750315371 1:1.1, full-wave Schottky.
CLK low = internal 420 kHz clock.

ISOLATED CAN FD



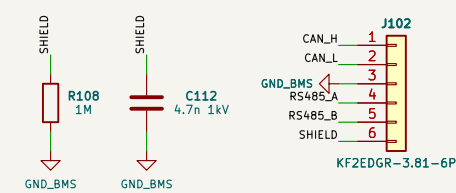
Split termination 2 x 60.4R + 4.7n, fitted by closing JP.

ISOLATED RS-485



120R termination fitted by closing JP.
Receiver is fail-safe (open/short/idle) inside ISO1410.

FIELD CONNECTOR (BMS)



Shield is RC-bonded to the Isolated ground.

Not bench-validated. PELV = controller side, BMS = field side.
Sheet 2
Schematics authored by Chinmoy Bhuyan
Barrier: 5 kVrms reinforced isolators; bias transformer 2500 VAC test (functional/basic)

Sheet: /01 Isolated CAN + RS-485/
File: 01_gateway.kicad_sch

Title: BMU-GW BMS gateway - Isolated CAN + RS-485

Size: A3 Date: 2026-10-04 Rev: B0
KiCad E.D.A. 10.0.6 Id: 2/2